

Mathematics for Machine Learning:

Homework 11

1. Find the Fourier transform of the function f .

$$(a) \ f(t) = \begin{cases} 1, & \text{if } |t| < 0.5 \\ 0.5, & \text{if } |t| = 0.5 \\ 0, & \text{if } |t| > 0.5. \end{cases}$$

$$(b) \ f(t) = \begin{cases} 0, & \text{if } t < 0 \\ e^{-t}, & \text{if } t \geq 0 \end{cases}.$$

2. Let $f(x) = x^2$, $x \in [-\pi, \pi]$.

(a) Find the Fourier series of f .

(b) Prove that f is equal to its Fourier series.

(c) Using the result of the previous point, prove that $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.

3. Find the DFT of the sequence $x_n = \delta_n + 0.9\delta_{n-6}$, $n = 0, 1, \dots, 8$.

4. Let $N \in \mathbb{N}$, $k_0 \in \{0, 1, \dots, N-1\}$ and $\varphi > 0$. Find the IDFT of the sequence $x_k = e^{i\varphi}\delta_{k-k_0} + e^{-i\varphi}\delta_{k-N+k_0}$, $k = 0, 1, \dots, N-1$.